Dietary microparticles implicated in Crohn's disease can impair macrophage phagocytic activity and act as adjuvants in the presence of bacterial st...

OBJECTIVE AND DESIGN: Western diets regularly expose the gastrointestinal tract (GI) to large quantities (> 10(12)/day) of man-made, submicron-sized, particles derived from food additives and excipients. These are taken up by M cells, accumulate in gut macrophages, and may influence the aetiology of inflammatory bowel diseases (IBD). MATERIALS: We investigated the effects of common dietary microparticles on the function of macrophages from healthy donors or active Crohn's disease (CD) patients. METHODS: Macrophages were incubated for 24 h with microparticles before being assayed for cytokine production and phagocytic activity. RESULTS: Microparticles alone were non-stimulatory but, in the presence of bacterial antigens such as LPS, they could act as adjuvants to induce potent cytokine responses. Uptake of high concentrations of microparticles also impaired macrophage phagocytic capacity - but not their ability - to take up 2microM fluorescent beads. CONCLUSIONS: While dietary microparticles alone have limited effects on basic macrophage functions, their ability to act as adjuvants could aggravate ongoing inflammatory responses towards bacterial antigens in the GI tract.